MOBILE DRILL BRAIN

Calibration Technician Training Module 001

Role, Purpose, and Workflow Overview

Document type	Workspace 006-B technician training module
Module	001 - Role, Purpose, and Workflow Overview
Version	v0.3 SME-Cleanup Draft
Prepared	July 2026
Source basis	Centerpiece / Master Repository v0.2; Calibration Service Module Extraction Worksheet v0.1; Module 002 v0.2 SME equipment inputs; SPT Introduction Draft v6; Chapter 2 Field Lessons; Calibration Data Collection Cheat Sheet
Training lane	Calibration Technician Training - not customer prospecting
Status	Draft for review; SME-needed section updated after Module 002; not a final field SOP or analyzer manufacturer manual

Locked module purpose
Teach a new calibration technician what the calibration role is, why the work matters, how a calibration job flows at a high level, what information must be captured, what must not be promised, and when to escalate. This module prepares the trainee for later equipment, setup, measurement, troubleshooting, and report-handoff modules.

1. Source basis and training boundary

This v0.2 module is grounded in two current Mobile Drill Brain sources:

- Centerpiece / Master Repository v0.2 - repository philosophy, HSA workflow, drill rod system, Split Spoon/SPT context, and compatibility-thinking foundation.
- Calibration Service Module Extraction Worksheet v0.1 - calibration objects, rig/hammer fields, event scheduling fields, field workflow checklist, measurement fields, report status workflow, escalation rules, technician training fields, and do-not-promise rules.

Training boundary
This module is technician-facing workflow training. It is not a sales prospecting module, not a customer-history report, not a quote/pricing guide, and not a substitute for lead-trainer approval or analyzer manufacturer instructions.

2. Learning outcomes

After completing Module 001, the trainee should be able to:

1. Explain the purpose of automatic hammer calibration in plain language.
2. Describe the calibration technician's responsibility before, during, and after a calibration event.
3. Recognize the basic job flow from scheduling handoff through report handoff.
4. Identify the core information that must be captured for the rig, hammer, rod setup, measurement setup, data collection, and field notes.
5. Explain why questionable data cannot be accepted just to finish faster.
6. State what the technician must not promise to the customer.
7. Recognize when field issues should be documented, rerun, escalated, or handed off to report builders/service/engineering.

3. Why calibration matters

Automatic hammer calibration supports measurement, documentation, compliance, and customer confidence. It is not merely paperwork. In SPT work, the hammer, drill rod, anvil, rig setup, and testing conditions affect the energy delivered through the rod string to the sampler. Customers often need calibration information because project owners, DOT/government work, engineering firms, internal policies, or bid requirements may require evidence that hammer performance has been measured and documented.

For the technician, the job is to collect reliable, reviewable field data and preserve enough context that a report builder or reviewer can understand exactly what happened during the calibration event.

4. Calibration technician role

Role area	Technician responsibility	Training emphasis
Preparation	Arrive with the right job context, equipment expectations, rig/hammer information, rod setup needs, and safety mindset.	Do not assume the job will be quick or clean.
Customer/driller coordination	Meet with the driller, explain the process, set downtime expectations, and confirm how the hammer normally operates.	Clear communication prevents frustration and bad setup assumptions.
Setup and measurement	Create the rig-specific data file, input required variables, confirm connections, test sensor readings, and collect data at required intervals.	Accurate setup information matters before the first data point is trusted.
Data quality	Save, review, repeat, and troubleshoot as needed. Do not trust data until it has been reviewed.	Questionable data is a stop-and-check signal, not a reason to rush.
Documentation	Record field obscurities, site conditions, mechanical issues, odd behavior, reruns, and anything the report builder will need.	Another person may need to build the report from the technician's notes.
Escalation	Escalate when repeated data problems, refusal, mechanical failure, or unexplained issues remain after normal checks.	Do not overdiagnose or promise fixes beyond authority.

5. High-level calibration job workflow

Later modules will teach each phase in more detail. Module 001 introduces the complete job flow so a new tech understands how the pieces connect.

Phase	What happens	Why it matters
1. Scheduling handoff	Date, location, rig count, contact info, rod type, tech assignment, and logistics are confirmed before the event is treated as ready.	Calibration cannot be promised until people, travel, rig count, and location are known.
2. Arrival and customer check-in	Tech meets the driller/customer contact and explains the calibration process and expected downtime.	The customer needs to understand this is measurement and documentation, not production drilling.
3. Rig/hammer setup review	Confirm normal hammer operating RPM, connection condition, anvil/rod setup, and site readiness.	Normal operating setup helps produce meaningful calibration data.
4. Drill to starting depth	Customer/driller advances to the starting depth. Actual site conditions are documented when they differ from the plan.	Stable collection conditions and real field notes matter for report credibility.
5. File creation and variable entry	Create a rig-specific data file and enter required measurements and variables such as rod type, LP, LE, drop height, and slug weight.	Bad inputs can make good field work unusable.
6. Sensor and hardware check	Confirm analyzer hardware is reading sensor signals before collection.	Sensor problems should be caught before relying on data.
7. Data collection	Collect energy-transfer, BPM, max/min energy, deviation/wave behavior, and	The goal is reviewable, reportable data, not simply completing a sequence.

	related outputs at required intervals.	
8. Save and review	Stop the sequence, save the run, and review the data before moving on.	Do not trust data until reviewed.
9. Repeat and troubleshoot	Repeat at required intervals; if data seems off, check setup and rerun or escalate.	Four good samples are needed for report; more may be collected if needed.
10. Report handoff	Save files, document obscurities, provide photos/notes if needed, and move the event into report drafting/review workflow.	Report builders cannot complete the package if the field record is incomplete.

6. Core information the technician must capture

The technician does not need to memorize every database field on day one, but they must understand the information categories that make a calibration event usable.

Category	Examples	Why it matters
Customer/event	Company, contact, location, date, scheduled location type, rig count, whether rigs are all in one location.	Supports scheduling, report identity, customer follow-up, and logistics review.
Rig profile	Rig manufacturer, model, serial number, known mechanical issues, last relevant service/parts replaced.	Rig serial and context are needed before final report completion.
Hammer profile	Hammer manufacturer, serial if available, install type, known issues, hammer operating RPM.	Hammer identity and behavior affect documentation and interpretation.
Rod/anvil setup	Rod type, anvil compatibility, flush/snug rod connections, rod-to-anvil context.	Incorrect or loose setup can create unwanted vibration or bad readings.
Measurement inputs	LP, LE, starting depth, drop height, slug weight, instrumented rod fields, analyzer-required values.	Data quality depends on accurate inputs.
Collection outputs	Average energy transfer, BPM, max/min energy, deviation, ETR, N, N60 when included.	These support the report summary and customer-facing documentation.
Field obscurities	Refusal, shallow depth, unstable conditions, bad hits, reruns, sensor issues, mechanical failures, odd customer setup, anything unusual.	Required when the report builder is not the same person who collected the data.

7. Measurement concepts introduced in Module 001

This module only introduces the terms. Later modules teach how the technician confirms, enters, reviews, and troubleshoots them.

Term	Plain-English meaning	Training note
Rod type	Rod family used for the calibration, such as AWJ or NWJ.	Rod setup must be correct.
LP	Length of rod from ground surface to bottom of borehole/spoon contact.	Rod above ground does not count.
LE	Distance from bottom of borehole to the first sensor on the instrumented rod.	Often around 3.5 ft, but must always be confirmed.
Starting depth	First intended collection depth.	Document if site conditions require a shallower or different setup.
Drop height	Expected hammer drop height.	Standard SPT hammer drop height is 30 inches.
Slug weight	Hammer weight used for the calibration context.	Usually 140 lb unless special case.
Average energy transfer	Average transferred energy for a collection.	Core calibration output.
BPM	Blows per minute.	Used for interpretation and report summary.
ETR	Energy Transfer Ratio - rated.	Common report output.
N / N60	SPT N-value and energy-corrected N-value when included.	Important report/customer-facing values when applicable.

8. Do-not-promise and do-not-assume rules

Core rule
A calibration technician should communicate clearly, but must not overpromise results, schedule outcomes, report delivery, field fixes, or pass/fail expectations.

Category	Do not promise / assume	Correct behavior
Scheduling	Do not promise a date until tech availability, travel, location, rig count, and logistics are confirmed.	Route scheduling commitments through the approved scheduling process.
Fleet days	Do not promise multiple rigs can be completed in one day without confirming location, site drilling capability, and likely troubleshooting risk.	Ask the right questions and build buffer time.
Jobsite risk	Do not assume jobsite calibrations will go smoothly or quickly.	Explain that field variables can create delays.
Results	Do not promise a hammer will pass or predict a specific energy percentage before testing.	Collect and review data first.
Data quality	Do not accept questionable data just to finish faster.	Check setup, rerun, document, or escalate.
Reports	Do not promise same-day delivery unless the report builder has confirmed it.	Standard guarantee is within 3 days unless confirmed otherwise.
Communication	Do not position calibration as just paperwork.	Describe it as measurement, documentation, compliance, and customer-confidence support.

9. Escalation and troubleshooting awareness

A new technician should recognize the first-level escalation categories even before learning detailed troubleshooting. Module 006 will cover data quality and troubleshooting in depth.

Condition	Required action	Do not assume
Data seems off at first depth	Move to the next testing depth and repeat after checking setup.	Do not assume hammer failure.
Data remains off after repeated collection	Troubleshoot hardware/software, rod connections, drop height, RPM, and mechanical issues.	Do not accept questionable data to finish faster.
Refusal or spoon does not advance enough	Treat as possible ground condition/data collection issue.	Do not automatically blame the hammer.
Jobsite mechanical failure or broken/lost tooling	Document the issue, coordinate service/engineering as needed, and reset expectations.	Do not promise field rescue will be possible.
Troubleshooting hierarchy followed and no obvious cause found	Escalate to engineering, drilling manager, or service lead.	Do not overdiagnose in the field or over the phone.

10. Report handoff awareness

Module 001 does not train the new tech to build the final report. It teaches why field documentation must support the report builder and review process.

Status	Meaning	Exit criteria / training note
Scheduled	Calibration event is booked but not completed.	Date, location, rig count, tech assignment, rod type, and contact info confirmed.
Data Collection Complete	Field/analyzer data has been collected.	Data files saved; obscurities documented.
Report Drafting	Report builder is assembling the formal package.	Metadata, analyzer outputs, summary pages, and certificates are available.

Needs Clarification	Report builder cannot complete package due to missing or unclear field information.	Field tech provides notes, photos, explanation, or corrected data.
Ready for Review	Report package is complete and ready for internal review/sign-off.	No missing pages or unresolved field issues.
Delivered	Final report delivered to customer.	PDF saved and delivery date recorded.

11. Training scenario

Scenario
A customer has one rig ready for calibration at their yard. The driller is experienced but in a hurry. The hammer is operating, but the first data collection looks inconsistent. The customer asks whether the hammer is failing and whether you can still get the report today.

Trainee response should include:

- Do not say the hammer is failing from the first bad-looking run.
- Check setup first: rod connections, anvil condition/fit, sensor readings, drop height/RPM context, and other measurement inputs.
- Move to another depth and repeat as appropriate.
- Document the issue and any reruns.
- Do not promise same-day report delivery unless the report builder has confirmed it.
- Escalate if repeated data remains questionable after normal checks.

12. Knowledge check

#	Question	Answer key
1	Why should a tech avoid describing calibration as just paperwork?	Because calibration is measurement, documentation, compliance support, and customer-confidence support.
2	What should a tech do if data seems off at the first depth?	Check setup and repeat/move to another depth; do not assume hammer failure.
3	What two measurements introduced in this module must be confirmed instead of guessed?	LP and LE.
4	Can the tech promise the hammer will pass before testing?	No.
5	Why must field obscurities be documented?	The report builder may need those notes to understand and complete the report.
6	What is the standard report-delivery expectation listed in the source?	Within 3 days of data collection unless report builder confirms otherwise.
7	What is the correct action if questionable data remains after repeated collection and setup checks?	Troubleshoot through the hierarchy and escalate to the proper lead/engineering/service path.

13. Practical signoff checklist for Module 001

Signoff item	Pass standard	Trainer initials/date
Explains purpose	Trainee can explain why calibration matters without calling it paperwork.	
Describes role	Trainee can describe preparation, setup, data collection, documentation, and escalation responsibilities.	
Identifies key captured information	Trainee can list customer/event info, rig/hammer profile, rod/anvil setup, measurement inputs, data outputs, and field obscurities.	
Applies do-not-promise rules	Trainee does not promise pass/fail, energy percentage, schedule, or same-day report without confirmation.	
Handles scenario	Trainee responds correctly to inconsistent first-depth data and customer pressure.	

14. SME-needed items and next module inputs

This v0.3 cleanup updates Module 001 after Module 002 and the new calibration training sources supplied after the v0.2 draft. Items that were open only because Module 002 had not yet been built are now marked resolved. Items needed for later modules remain listed so they are not lost.

Resolved since Module 001 v0.2:

- Equipment/analyzer kit inventory and manufacturer kit image were supplied and incorporated into Module 002 v0.2.
- Baseline PPE has been confirmed as hardhat and steel-toe boots, with additional PPE controlled by customer/jobsite requirements.
- Module 002 now covers the Pile Dynamics Inc. SPT Analyzer, pigtail/cable path, 19-pin twist-lock connectors, AWJ/NWJ instrumented rods, accelerometers, strain gage bridges, battery/storage/tooling items, dielectric-grease connector care, and equipment-damage escalation.
- SPT Introduction Draft v6 and Chapter 2 Field Lessons have been supplied as source material for later theory, measurement, and data-quality modules.
- Calibration Data Collection Cheat Sheet and SME clarification have been supplied as source material for the later SPT Analyzer software workflow module.

Still needed for later modules:

- Exact analyzer button-by-button procedure should be converted into a formal training module from the cheat sheet plus SME clarification, rather than treated as a manufacturer manual replacement.
- Lead trainer decision on minimum observed calibrations before solo approval.
- Final Mobile Drill report template package and required report-builder/reviewer responsibilities.
- Current escalation contact path and authority levels for questionable data, damaged equipment, mechanical failure, or report issues.
- Any site-specific PPE/customer safety exceptions that should be called out beyond the Mobile Drill baseline.

Next module direction:

Module 003 should cover SPT Fundamentals, N-value, N60, ASTM D1586 context, and why calibration matters, using the SPT Introduction Draft v6 as its primary source. Module 004 should later convert the calibration cheat sheet and SME clarification into a structured SPT Analyzer software workflow module.

Training track status
Module 002 v0.2 is complete.
Next new module: Module 003 - SPT Fundamentals, N-value, and N60.
Later procedure module: Module 004 - Field Setup and SPT Analyzer Software Workflow.

Revision history

Version	Status	Change
v0.1	Draft	Initial Module 001 created from Centerpiece source with calibration gaps marked SME-needed.
v0.2	Source-backed draft	Updated with Calibration Service Module Extraction Worksheet v0.1: field workflow, measurement fields, report workflow, escalation rules, training fields, and do-not-promise rules.
v0.3	SME-cleanup draft	Updated SME-needed section after Module 002 v0.2 and new source uploads. Marked equipment kit and baseline PPE items resolved, registered SPT Introduction Draft v6, Chapter 2 Field Lessons, and Calibration Data Collection Cheat Sheet as inputs for Modules 003-006, and clarified next-module direction without converting Module 001 into a procedure manual.